

Deep Learning for Time Series

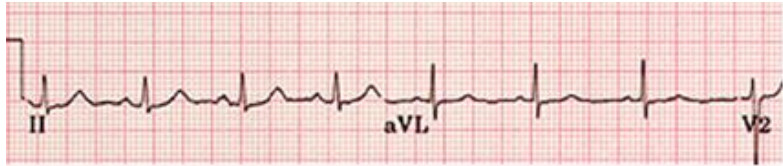
Session 4b: Time Series Foundation Models

Romain Tavenard

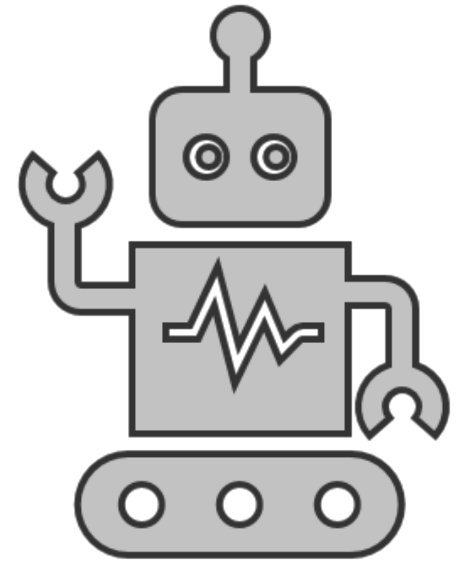
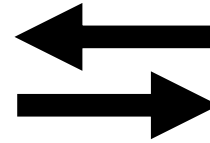
NB: Some figures in this slide deck are borrowed from Ievgen Redko's (great) course on time series forecasting and classification

Traditional Machine Learning

Normal ECG



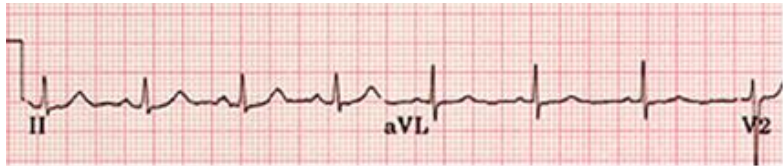
Abnormal ECG



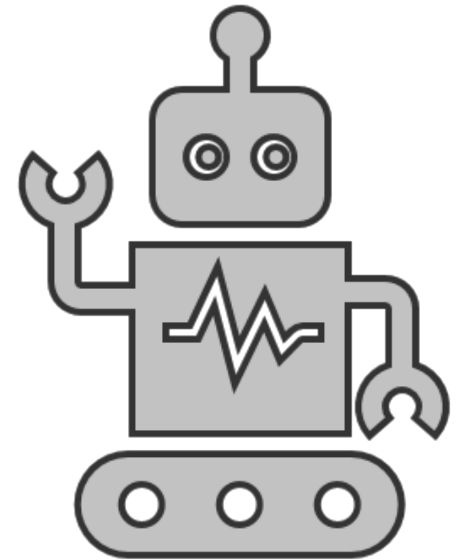
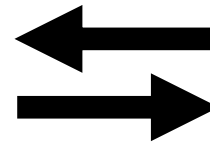
- Traditional ML pipeline
 1. Collect training data
 2. Train a new model
 3. Deploy it

Traditional Machine Learning

Normal ECG



Abnormal ECG



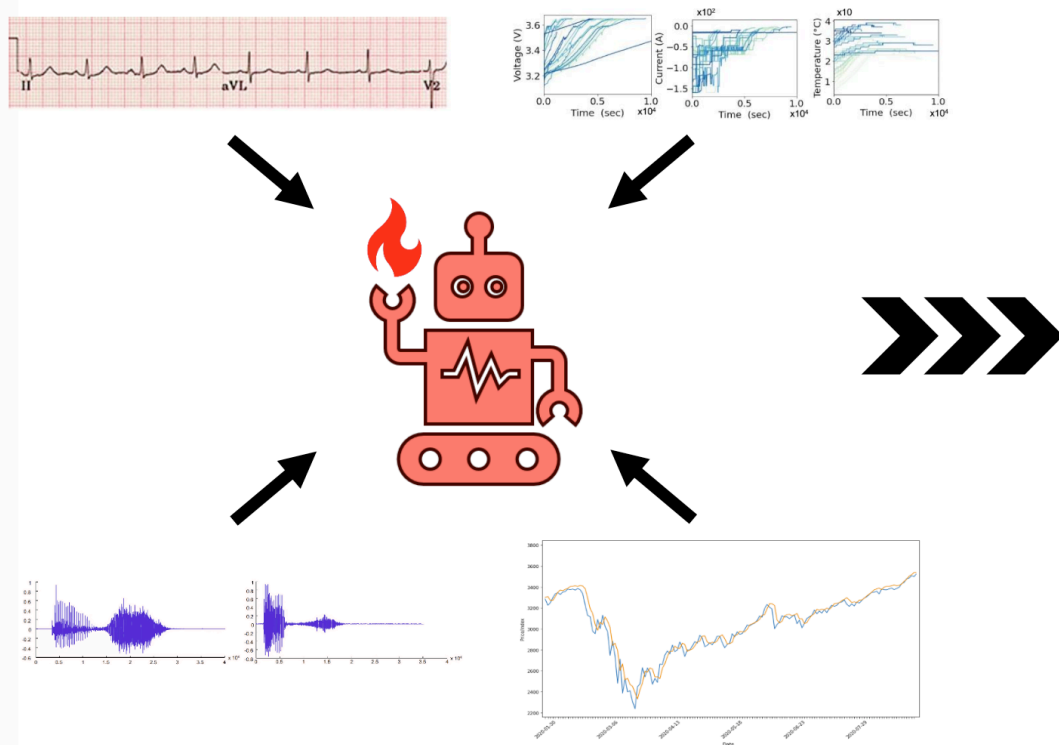
- Traditional ML pipeline
 1. Collect training data ← Issue 1: requires large training set
 2. Train a new model ← Issue 2: a new model for each task
 3. Deploy it

Success of pre-training

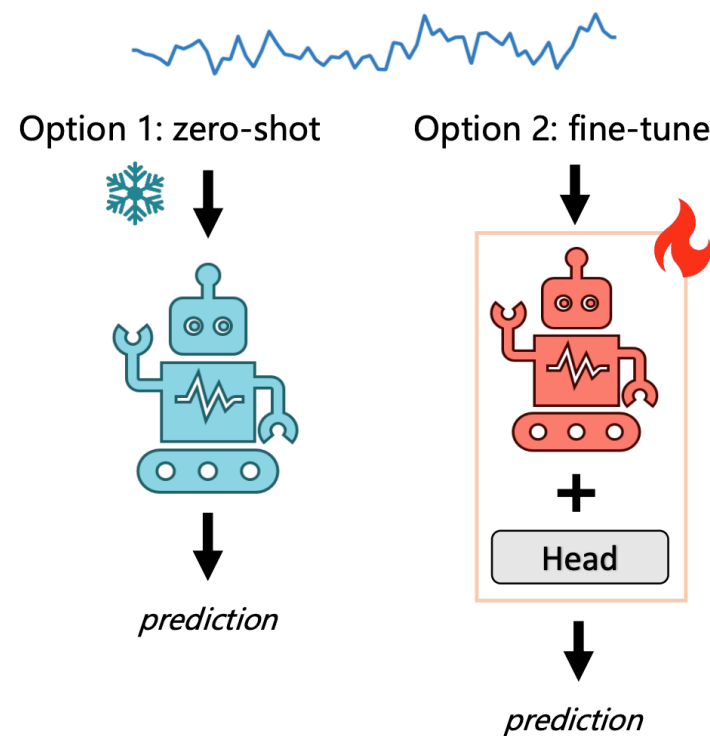
- In Computer Vision: ImageNet pre-training is everywhere
- In text: multi-task LLMs
- In time series
 - Is there an ImageNet?
 - Do we have competitive task-adaptive models?

Time Series Foundation Models (TSFMs)

Step 1: Pre-training



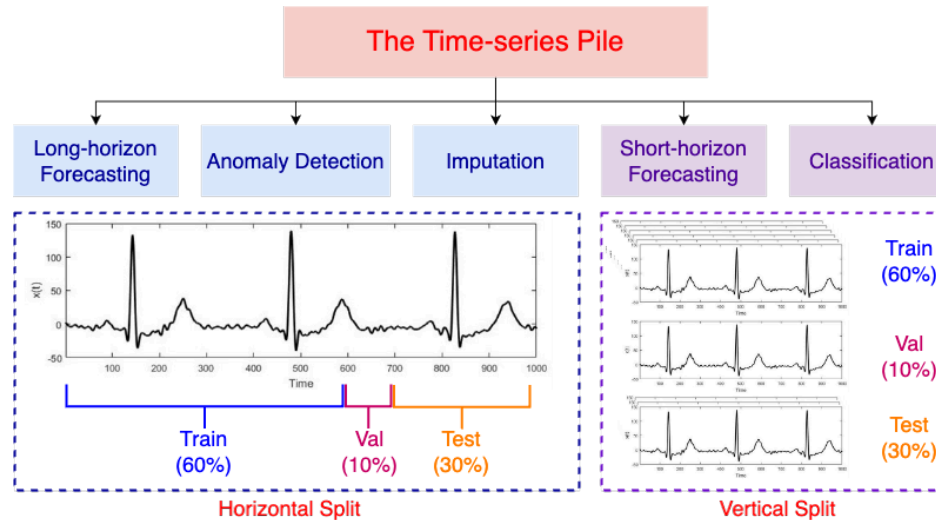
Step 2: Use on New Tasks



Step 1: Pre-training

- LOTSA (MOIRAI)
 - 27.7B observations
 - 4M time series
 - LOTSAMini exists: $< 1\%$ of LOTS
- TimeSeries-PILE (MOMENT)
 - 1.1B observations
 - 13M time series
- Time300B
 - 300B observations
 - 48M time series
- TimePFN's synthetic data generators
 - used 300-600M observations for TimePFN pre-training

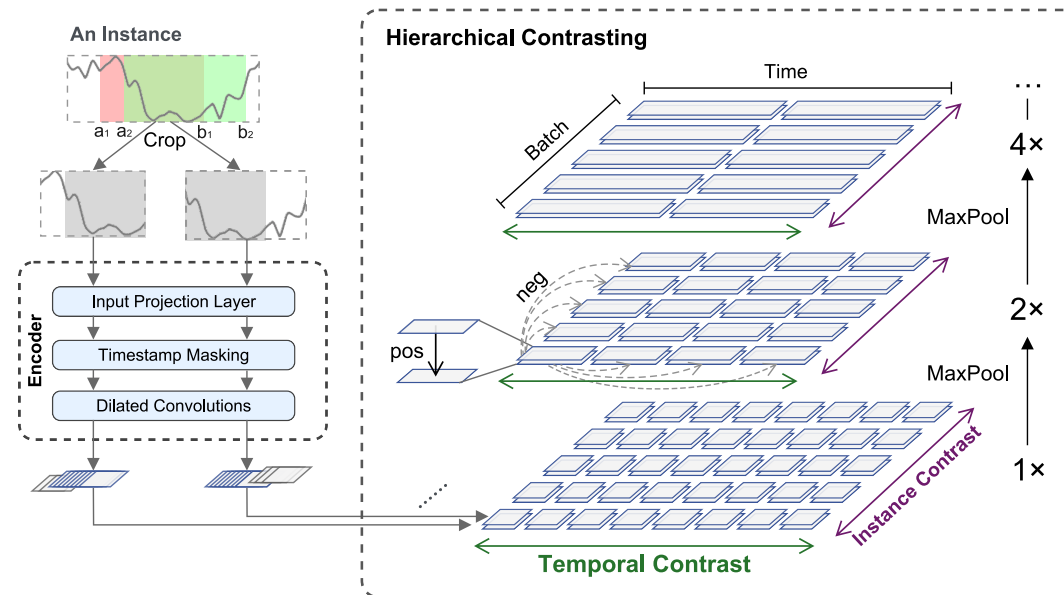
- Pre-training across tasks



Source: “MOMENT: A Family of Open Time-series Foundation Models”,
ICML’24

- Semi-Supervised Learning
 - eg. MOMENT (masking), TS2Vec (contrastive)
- Use pre-trained LLM
 - eg. GPT4TS

- Self-supervised contrastive learning for time series
 - Contrast positives: same series, different segments
 - Contrast negatives: different time series
- Learns multi-scale (hierarchical) representations
- Temporal encoder (CNN-based), no decoder
- Outputs fixed-length embeddings for downstream tasks

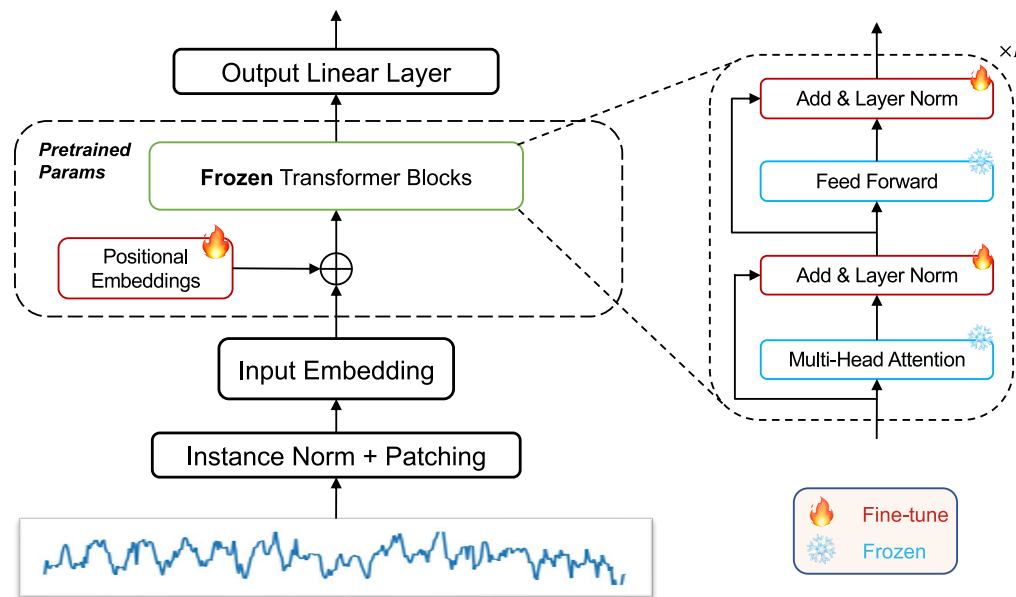


Source: "TS2Vec: Towards Universal Representation of Time Series", AAAI'22

- First strong task-agnostic TS representation method
- Validated pretrain once → reuse everywhere paradigm
- Shifted SSL from reconstruction to semantic contrastive learning
- Strong zero-shot / linear-probe baseline

General idea: make use of LLM backbone

1. Normalization + Patching (à la PatchTST)
2. Projection to LLM embedding dimension
3. Fine-tuning of positional embeddings + layer normalization, keeping others frozen



Source: “One Fits All: Power General Time Series Analysis by Pretrained LM”,
NeurIPS’23

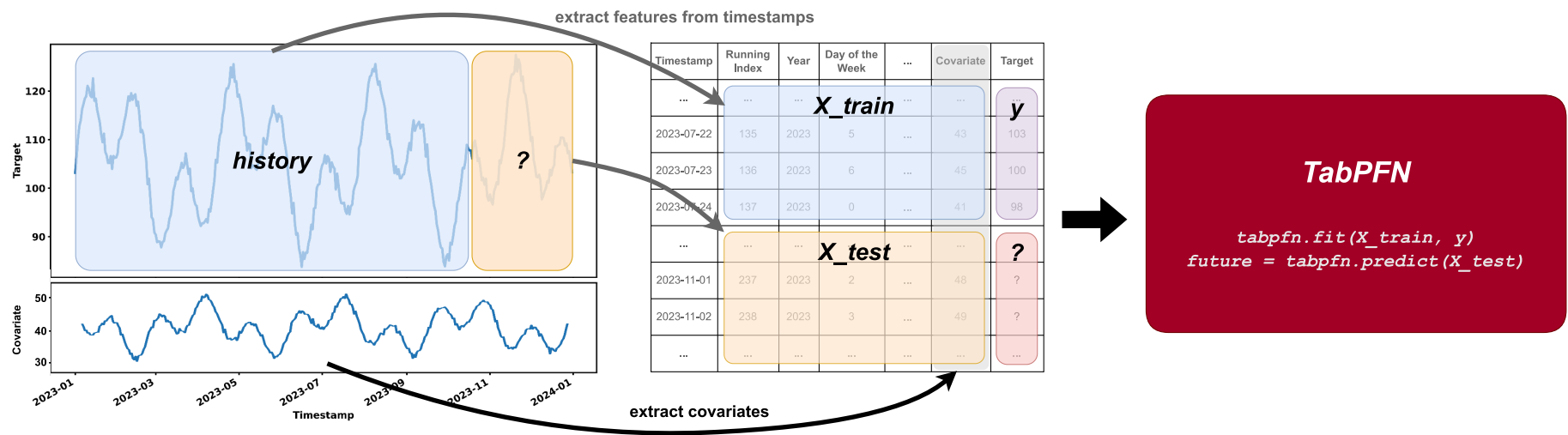
Step 2: Adapting

Extract patches at different scales to catch multi-resolution information



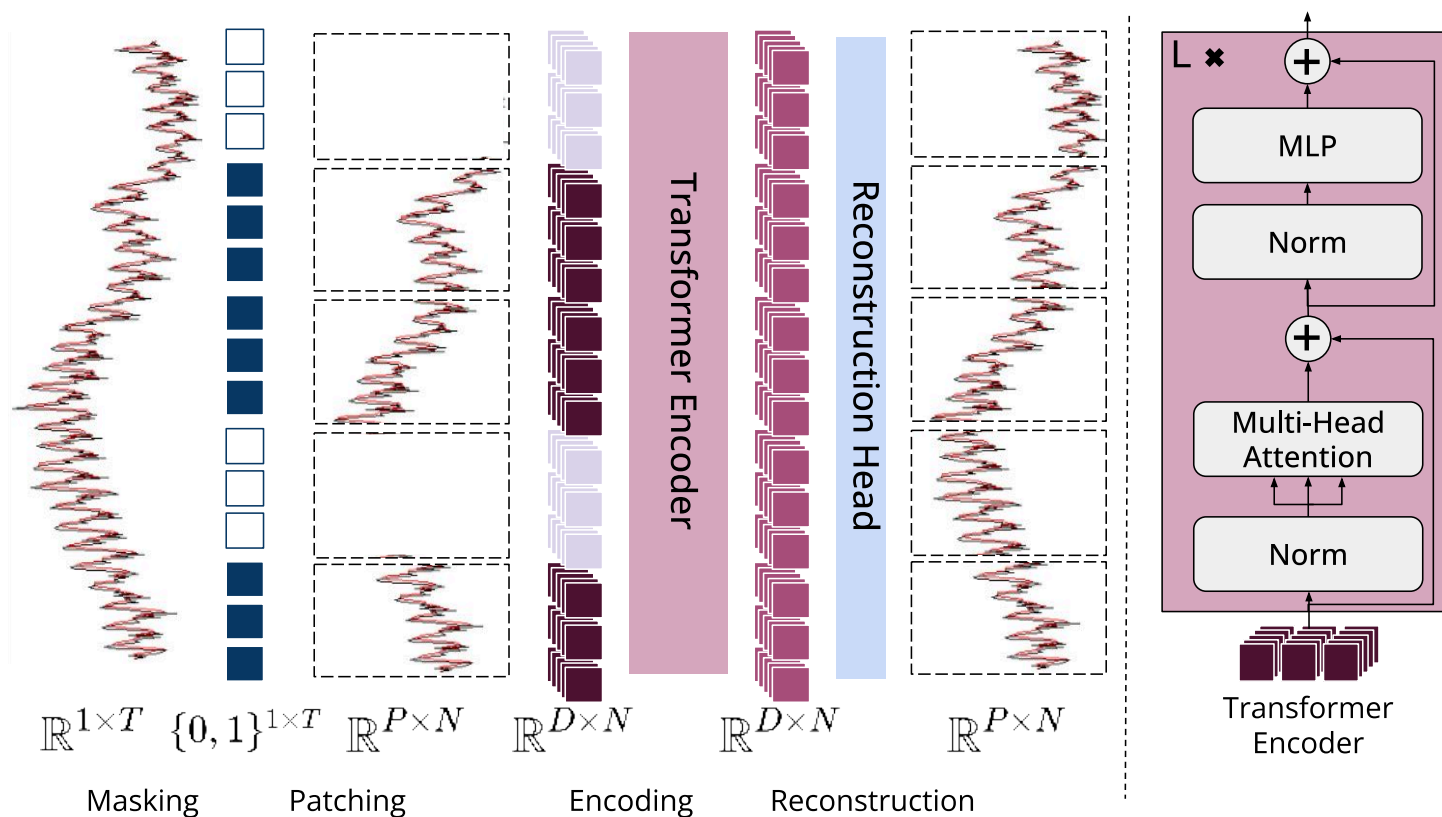
Source: “Unified Training of Universal Time Series Forecasting Transformers”,
ICML’24

- TabPFN is pre-trained on tabular data alone
- Builds basic TS features (calendar, seasonal, ...)
- No TS prior at all!



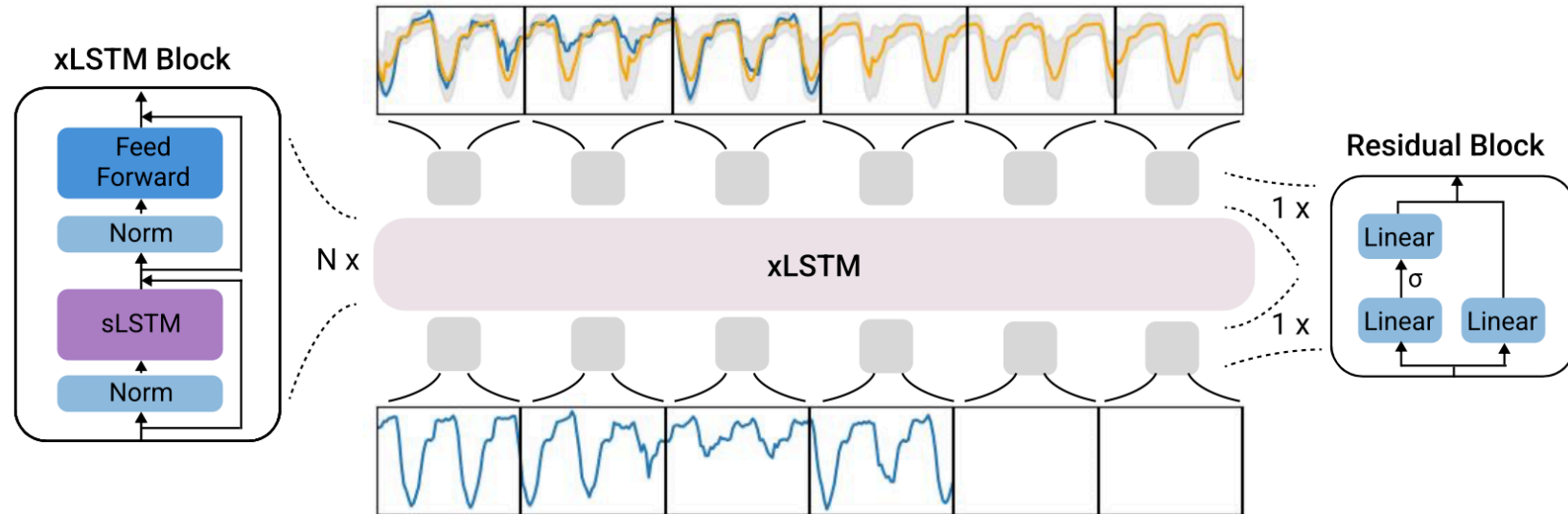
Source: “From Tables to Time: Extending TabPFN-v2 to Time Series Forecasting”, ArXiv’26

Encoder-only model: can be used for any downstream task (classification, forecasting, imputation etc)



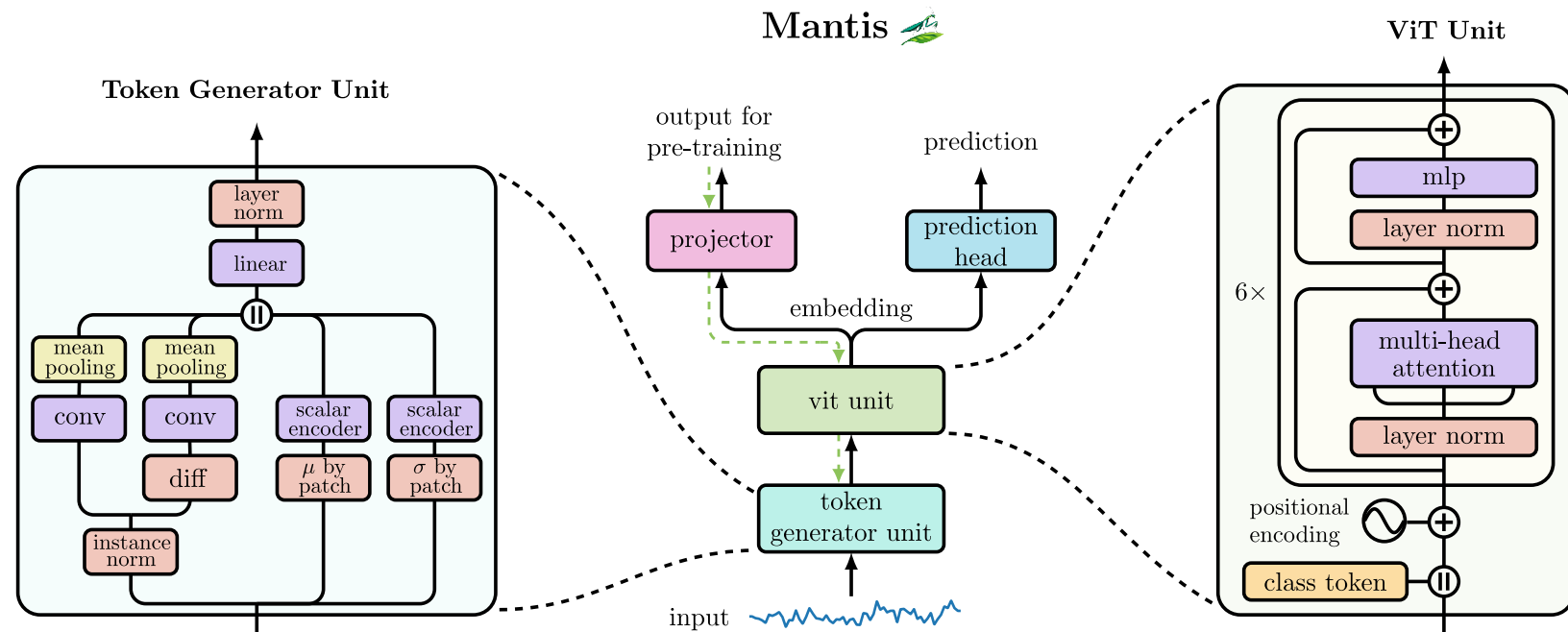
Source: "MOMENT: A Family of Open Time-series Foundation Models", ICML'24

- Masking-based pre-training
- Simple xLSTM architecture (only sLSTM blocks)



Source: “TiRex: Zero-Shot Forecasting Across Long and Short Horizons with Enhanced In-Context Learning”, NeurIPS’25

- Contrastive pre-training
- Scalar embedding module for preserving basic statistics
- Strong zero-shot and fine-tuning performance for TSC



Source: "Mantis: Lightweight Calibrated Foundation Model for User-Friendly Time Series Classification", ICML workshops'25

Step 3: Evaluating

GIFTEval — Benchmarking Time-Series Foundation Models

Step 3: Evaluating

- Unified evaluation benchmark for TSFMs
- Focuses on forecasting
- Diverse datasets across domains and time scales
 - short / long term
 - univariate / multivariate
- <https://huggingface.co/spaces/Salesforce/GIFT-Eval>